Quality Management

8 — Digital images: pixels, grey levels, colour and files

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Today

From an image of light to a matrix of numbers

Lecture 7 ended with an image of light on the sensor. Today it becomes **numbers** — and numbers are what make an image something you can measure, compare, store and decide on.

0 **Recap and why**
lecture 7 in two slides; why digital

1 **Pixels**
sampling, aliasing, resolution

2 **Grey levels**
quantisation, bit depth, saturation

3 **Colour**
channels, colour spaces, measuring colour

4 **Files and records**
formats, compression, images as evidence

From lecture 7: forming the image

Recap, 1 of 2

Light and colour

Visible light: 380–700 nm. An object's colour depends on the object and on the light. The eye has three kinds of cone, so three channels are enough to reproduce a colour.

Image formation

Pinhole and lens: similar triangles. $1/f = 1/u + 1/v$, magnification $m = v/u$. Field of view $W = S(u - f)/f$.

Errors of the optics

Distortion, vignetting, chromatic aberration, perspective. Corrected by calibration on known targets, or avoided with telecentric lenses.

From lecture 7: the settings and their price

Recap, 2 of 2

Setting	Gives	Costs
Focal length & distance	the field of view; less perspective from far	detail per part of the field
Aperture (larger *N*)	more depth of field	light (half per stop); diffraction
Exposure time (shorter)	no motion blur	light
Gain (ISO)	brightness	noise; earlier saturation

The rule

Decide from what must be seen; choose the smallest *N* that gives the depth; make up the light with lighting, not gain.

What the optics leave out is gone

Blur, saturation and missing depth cannot be repaired by any processing. The digital step that follows can only keep — or lose — what the optics delivered.

Digital images are everywhere

Most of them are never looked at by a person: they are measured by software

Phones
Several cameras each; faces unlock the phone; documents scanned into text.

Cars
Cameras keep the lane, read speed signs and brake for pedestrians.

Medicine
Digital X-ray, CT, endoscopy, microscopy of tissue samples.

Shops and logistics
Barcode scanners at the till; parcels sorted by reading their labels in flight.

Traffic and security
Number-plate recognition for tolls and access; speed cameras.

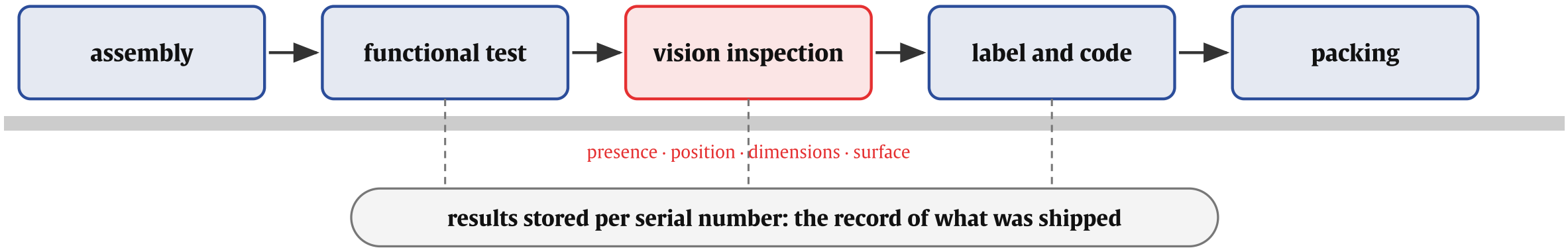
Sport
Goal-line technology and line calls from several synchronised cameras.

Science
Telescopes, microscopes, satellites: images measured, not just viewed.

Industry
On almost every production line — next two slides.

In quality management: end-of-line control

The last check before a product leaves the factory



- **End-of-line (EOL) control**: the stations at the end of an assembly line where **every unit** is tested before it is packed — functional tests (does it work?) and inspections (is it complete, correct, undamaged?).
- Cameras do the inspections: presence and position of components, dimensions, surface, labels and codes — in the seconds the line allows for each unit.
- The result of every check is stored with the unit's **serial number**: the record of what was shipped.

The e-scooter of lecture 6

Its end-of-line test was a test ride on a roller bench. A vision station next to it would check that the brake cables are routed, the screws are present, the frame is unscratched and the label matches the model — every scooter, not a sample.

What cameras check, industry by industry

The same few tasks, everywhere

Industry	Typical vision checks
Automotive	gap and flush between body panels · paint defects in an inspection tunnel · clips, connectors and screws present · labels and identification numbers
Electronics	automated optical inspection of circuit boards: every component present, the right way round, every solder joint formed
Pharmaceuticals	blister packs full and tablets unbroken · label and leaflet text verified · a unique code on every pack, read and recorded for traceability
Food and drink	fill level, cap and seal · foreign bodies · sorting fruit, grain and nuts by colour and shape
Metals, plastics, paper	surface defects on strip, film and sheet at full line speed · dimensions of machined and moulded parts
Packaging and logistics	print quality and position · seal integrity · codes read for every case and pallet

Five tasks behind all of them

presence · **position** · **dimension** · **surface** · **identification** (codes and text)

What makes them possible

Every one of these checks is a calculation on the numbers of a digital image — none of them could be done on film.

Why digital: analog and digital images compared

The same optics; a different kind of result

	Analog (film)	Digital
Result	a picture, to be looked at	**numbers**, to be computed with
Available	after chemical development: minutes to hours	at once: hundreds of images per second
Judged by	a person	an algorithm: the same answer for the same image
Inspection	samples	**every part**, at line speed
Response to light	non-linear, depends on the film and its development	linear and calibratable (lecture 9)
Copies	each copy loses quality	identical, as many as needed
Record	prints in an archive	files with time, settings and serial number attached
Cost per image	film and chemicals	almost nothing

Film still has its uses — very large formats, archival stability — but nothing in quality control needs it. The decisive difference is the first row: a digital image is **data**.

A digital image is a measurement

Why measurement science cares

Millions of measurements at once

Mega means million: a 5-megapixel camera has 5 million pixels, so one exposure gives **5 million light measurements**.

Calibrated, with an uncertainty

Pixels can be converted to millimetres and grey levels to light. The result has an uncertainty, as for any instrument.

Data for statistics

Results go straight into quality tools: the diameter of every part into a control chart (lecture 6).

Full-field methods — they measure a whole surface or volume at once, and exist only because images are numbers:

Method	Measures	Example
Digital image correlation	displacement and strain over a surface	a bracket under load
Particle image velocimetry	velocity of a fluid	the air flow around a fan
Thermography	surface temperature	a hot spot on a circuit board
Photogrammetry	3D coordinates	a car body checked against its CAD model

PART 1

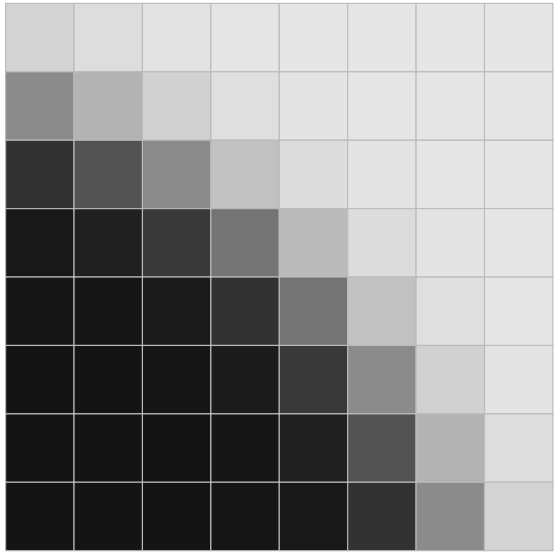
Pixels

The image as a matrix · Sampling · Aliasing · Resolution

A digital image is a matrix of numbers

What the camera stores is not a picture

what you see (zoomed in)



what is stored

212	221	227	229	230	230	230	230
139	179	209	223	228	230	230	230
50	83	139	193	220	228	230	230
24	32	57	116	187	220	228	230
21	22	27	50	116	193	223	229
20	20	21	27	57	139	209	227
20	20	20	22	32	83	179	221
20	20	20	21	24	50	139	212

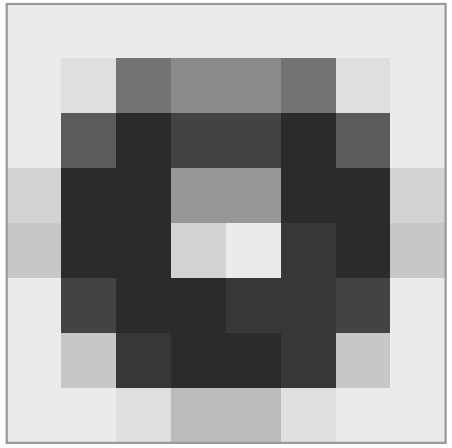
the edge of a dark hole in a bright part: about 20 inside, 230 outside

- The light on the sensor is continuous. The sensor is a grid of small light-sensitive cells — **pixels** (picture elements). Each collects the light falling on its area during the exposure and gives **one number**: its grey level.
- Two steps make an image digital: **sampling** — where, the grid of pixels — and **quantisation** — how much, in whole grey levels.
- A grey image is one matrix of M rows × N columns. Pixel (0, 0) is the top-left corner: row first, then column.
- Every inspection algorithm works on these numbers, never on the picture on the screen.

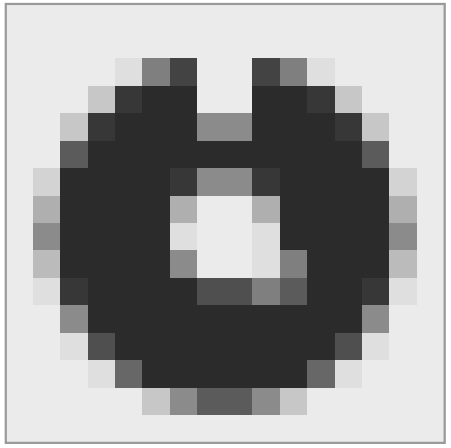
Sampling: how many pixels

The same part, recorded with three grids

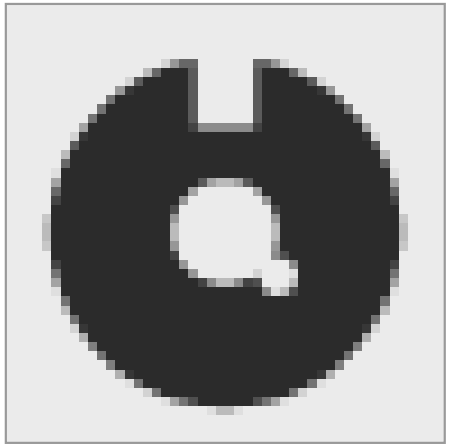
Definition · Sampling
Recording the continuous image only at a grid of points — the pixels — with **one value per pixel**. It decides *where* the image is known.



8 × 8 pixels



16 × 16 pixels



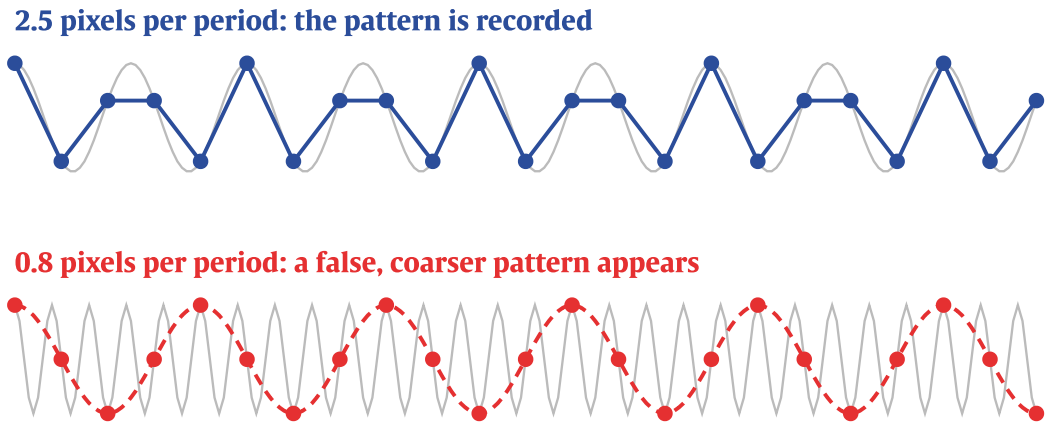
48 × 48 pixels

- Each pixel records the **average** light over its area: detail smaller than a pixel is lost — the small hole becomes a grey smudge.
- What matters is the pixel size **on the part**: 100 mm over 2448 pixels is 41 µm per pixel.

More pixels are not free
Twice the pixels across means four times the data, smaller pixels with less light each (lecture 9), and slower processing. Choose the grid from the smallest detail the task needs (slide 15), not from the largest sensor on sale.

Aliasing

When the grid of pixels is too coarse for the pattern



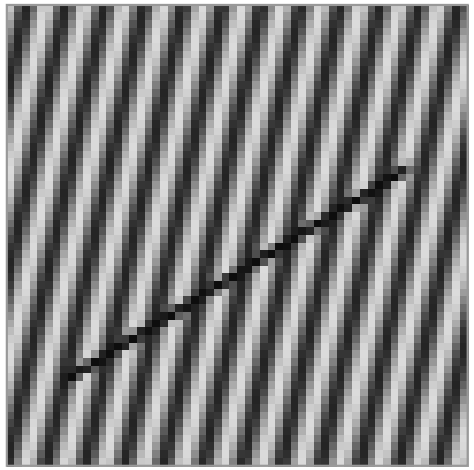
grey: the real pattern · dots: one value per pixel

Seen every day: moiré bands when filming a screen, striped shirts that shimmer on television, wheels that seem to turn backwards in films.

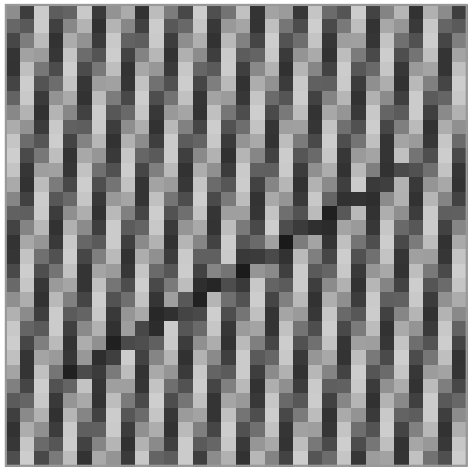
- **What it is** — a pattern finer than the pixels can follow is not lost: it is recorded as a **false, coarser pattern**.
- **Why it matters** — on woven fabric, wire mesh, machined grooves or screens, the false pattern can look like a defect, or hide one. Periodic features — a pitch, a thread — are measured wrong.
- **The rule** — the sampling theorem (Nyquist–Shannon): at least **two pixels per period** of the finest pattern in the scene.
- **How to avoid it** — more pixels on the part (smaller field, longer lens); or remove the finest detail before sampling: many cameras have an optical low-pass filter in front of the sensor.

How many pixels a pattern needs

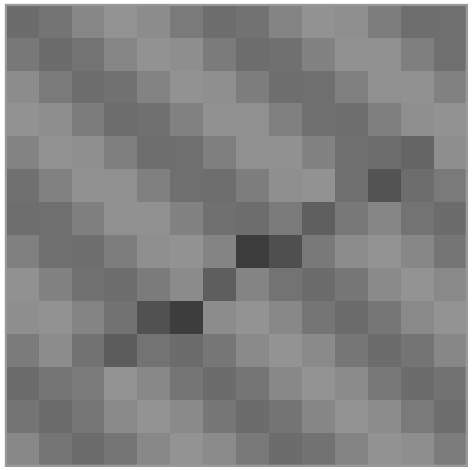
The same striped surface, with a scratch, recorded with three pixel counts



60 × 60 pixels · 5 per stripe



32 × 32 · 2.7 per stripe



14 × 14 · 1.2 per stripe

- **Plenty of pixels** (5 per stripe): the stripes and the scratch are both there.
- **Just enough** (2.7 per stripe): the stripes are recorded, but the thin scratch is already hard to see.
- **Too few** (1.2 per stripe): the image shows **wide stripes in the wrong direction** that are not on the part, and the scratch is reduced to a few dark dots.

For inspection

Count the pixels on the **finest thing you must see**: at least two per period of a pattern, two to three across a defect. Too few pixels do not just blur the image — they can invent structure that is not there.

Resolution: one word, five meanings

And how many pixels a task really needs

"Resolution" as…	Measured in	Example
pixel count	pixels, megapixels	2448 × 2048 = 5 MP
dots per inch	dpi (scanners, printers)	300 dpi = one dot every 85 µm
pixel on the object	µm per pixel	41 µm on a 100 mm field
optical resolution	smallest detail the lens separates	set by focus, aberrations, diffraction
measurement resolution	smallest change the result shows	can be finer than a pixel (lecture 12)

Task	Rule of thumb
find a defect	at least 2–3 pixels across it
read printed characters	about 20 pixels of character height
read a 2D code	at least 2–3 pixels per module, more is safer
measure an edge	a fraction of a pixel, with calibration

Rules of thumb from practice; the software's datasheet gives the exact figures.

PART 2

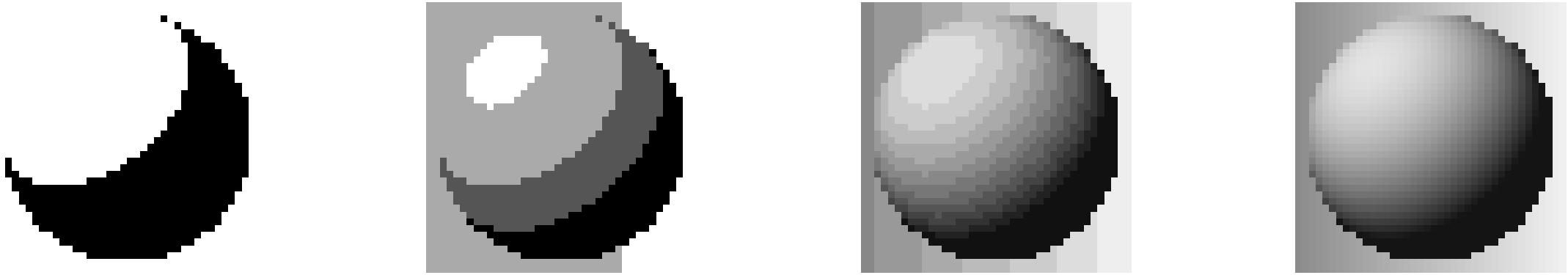
Grey levels

Quantisation · Bit depth · Rounding error · Saturation

Quantisation: how many grey levels

The same image with 2, 4, 16 and 256 levels

Definition · Quantisation
Rounding the light measured by each pixel to **one of a fixed number of levels**, stored as a whole number. It decides *how finely* brightness is known.



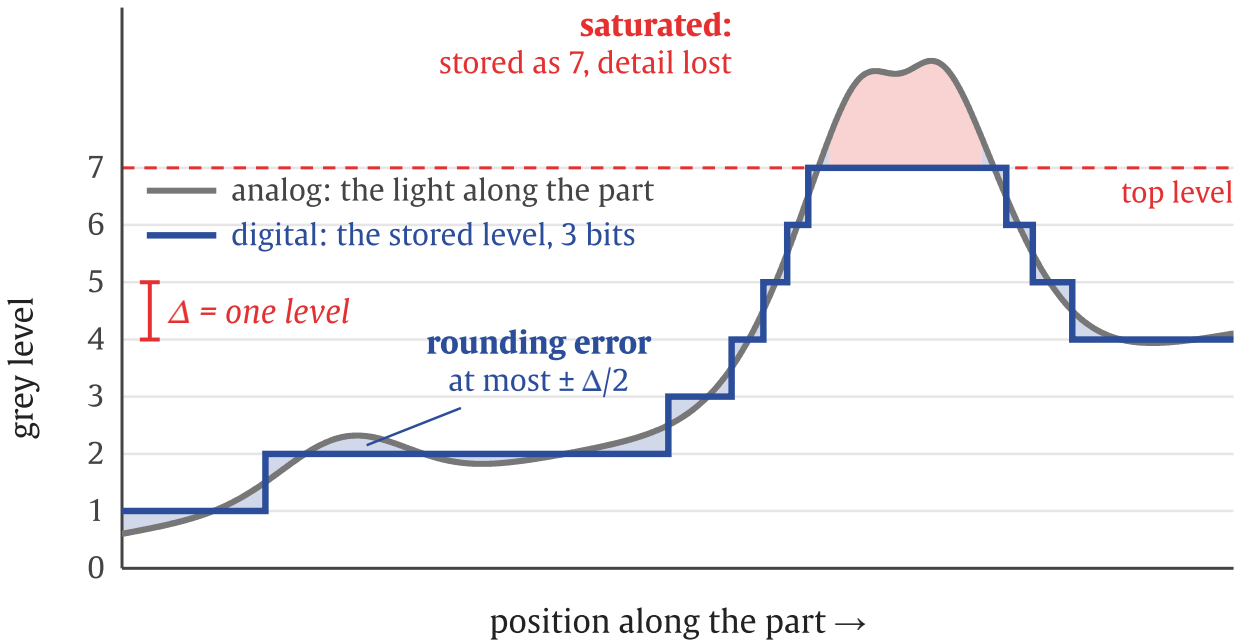
1 bit · 2 levels | **2 bits · 4 levels** | **4 bits · 16 levels** | **8 bits · 256 levels**

- With b bits there are $\mathbf{2^b}$ **levels**: 8 bits give 256.
- Too few levels and smooth shading turns into bands (*posterisation*), and small differences of brightness — a faint stain, a shallow dent — vanish into one level.
- 1 bit: black or white. That is what a threshold produces on purpose (lecture 11).

Bits	Levels	Typical use
8	256	screens, most images
10–12	1024–4096	machine vision
12–14	4096–16 384	camera raw files
16	65 536	scientific and medical

Rounding error and saturation

What quantisation adds to every value, and what it destroys at the top



Saturation

- **What it is** — light above the top level is stored as the top level (255 with 8 bits). The pixel is "full".
- **Why it matters** — the detail inside is lost for good; the edge next to it moves outwards; brightness and colour measured there are wrong.
- **How to see it** — a peak at the top value in the histogram (lecture 11).
- **How to avoid it** — expose for the brightest region you need, just below the top; reduce reflections (diffuse light, polarisers, lecture 10); lower the gain. More bits or several exposures only when the scene really needs them.

Rounding error

Every value is rounded to a step Δ: an error of up to ±Δ/2, with standard deviation $\Delta / \sqrt{12}$ — the same as a caliper's resolution in an uncertainty budget. Negligible when the noise is larger.

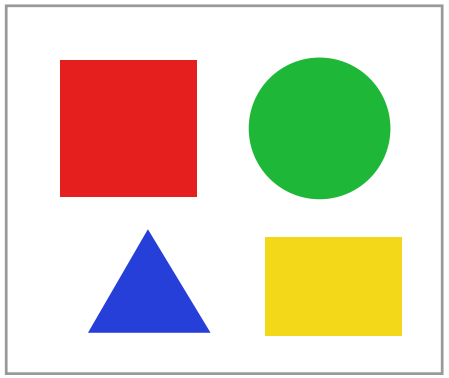
PART 3

Colour

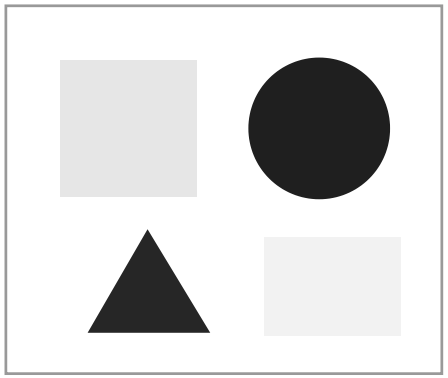
Three channels · Colour spaces · Measuring colour

A colour image is three grey images

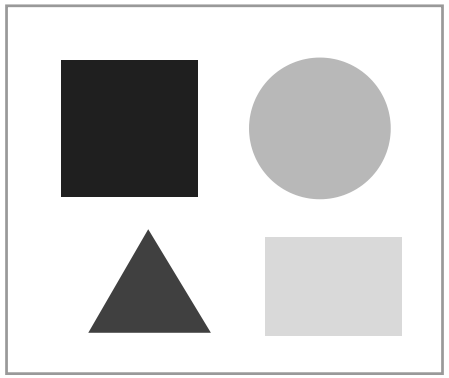
Built to imitate the eye's three kinds of cone (lecture 7)



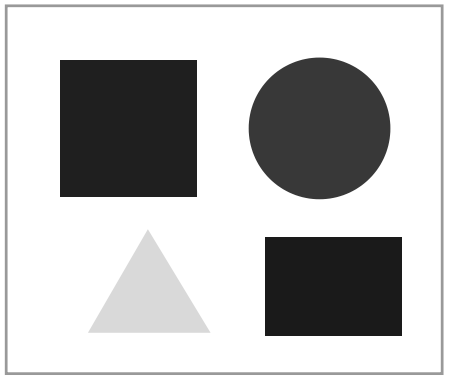
colour image



red channel



green channel



blue channel

- Three cone types: three numbers per point are enough.
- A colour camera records three grey images: **red**, **green**, **blue**.
- 3 matrices; 8 bits each: **16.7 million colours**.

Why it matters

Colour is something to check — paint shade, ripeness, wire colours. And a tool: a defect invisible in grey can stand out in one channel.

Colour spaces

The same colour, written in the coordinates that suit the job

Space	Coordinates	Used for
RGB	red, green, blue	cameras and screens: how colour is recorded
HSV	hue, saturation, value	colour checks under changing brightness: the hue barely moves when the light dims
CIELAB	lightness L*, a* (green–red), b* (blue–yellow)	**colour tolerances**: distances match what people see; its axes follow the eye's opponent pairs
CMYK	cyan, magenta, yellow, black	printing (lecture 7)
YUV	brightness + two colour signals	video and compression

Why change coordinates? RGB mixes brightness and colour: the same red part, under a dimmer lamp, has completely different RGB numbers. A space that separates them lets you check the colour while ignoring the light — or measure differences the way people see them.

A colour tolerance

A panel is accepted if its difference from the master is small:

$$\Delta E^*_{ab} = \sqrt{(\Delta L^{*2} + \Delta a^{*2} + \Delta b^{*2})}$$

People start to notice a difference of about 1–2.

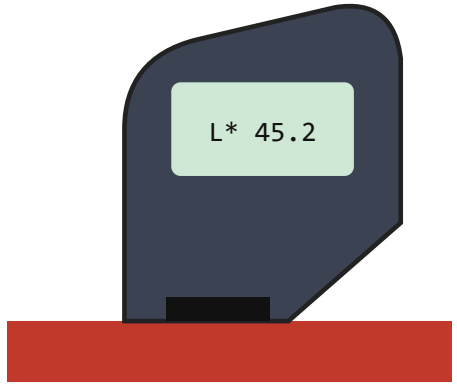
Measuring colour in quality control

From the light reflected by a part to a pass or fail on its colour

1 Light it and look
known lamp, fixed angle

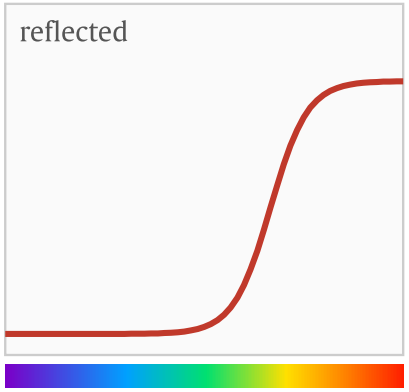
2 Light reflected,
wavelength by wavelength

3 Three numbers, compare
with the master, as ΔE



painted part

a spot of about 8 mm is measured



little blue and green, much red: the part is red

	L*	a*	b*
part	45.2	58.1	39.0
master	45.8	57.3	38.2

ΔE = 1.3
limit 2.0 → pass

computed for standard daylight and a standard observer

Spectrophotometer and colorimeter

The instruments of steps 1–3. A spectrophotometer measures the whole spectrum; a colorimeter uses three filters, like the eye. Both give L*, a*, b*: the reference for colour tolerances.

A colour camera

Every point at once, at line speed — but it only *measures* colour with a fixed light, fixed exposure and white balance, a reference colour target in the image, and tolerances set against the instrument.

PART 4

Files and records

File formats · Compression · Images as quality records

File formats

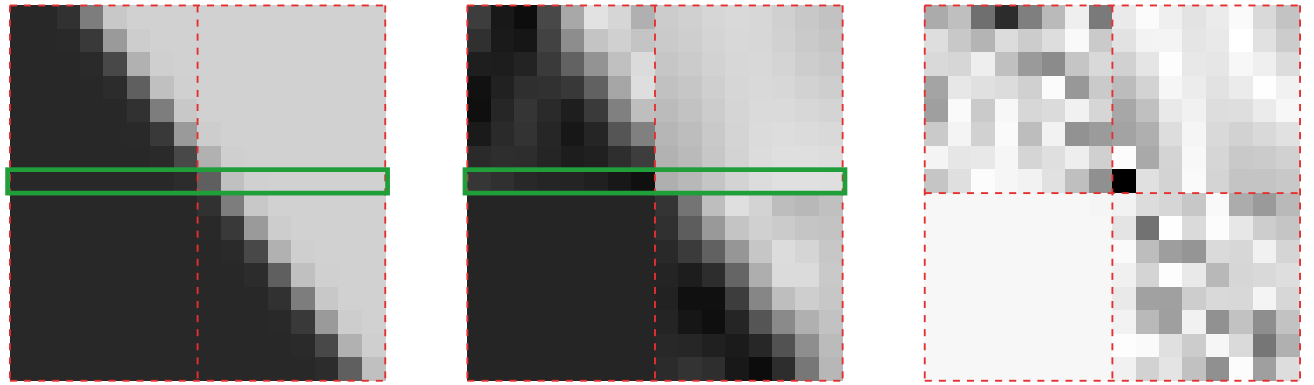
The file decides whether the numbers survive

Format	Compression	Bits per channel	Notes	For measurement?
RAW	none, or lossless	12–14	the sensor's own values, before any processing; camera-specific	**yes**
TIFF	none or lossless	8 or 16	the standard for industry and science; holds metadata	**yes**
PNG	lossless	8 or 16	widespread, compact	**yes**
BMP	none	8	simple, large	yes
JPEG	**lossy**	8	small files; fine detail thrown away	**no**
HEIC	lossy	8–10	phones' default	no
DICOM	lossless or lossy	up to 16	medical images, with patient and device data	with care
MP4 video	lossy, across frames	8–10	recording processes, not measuring them	no

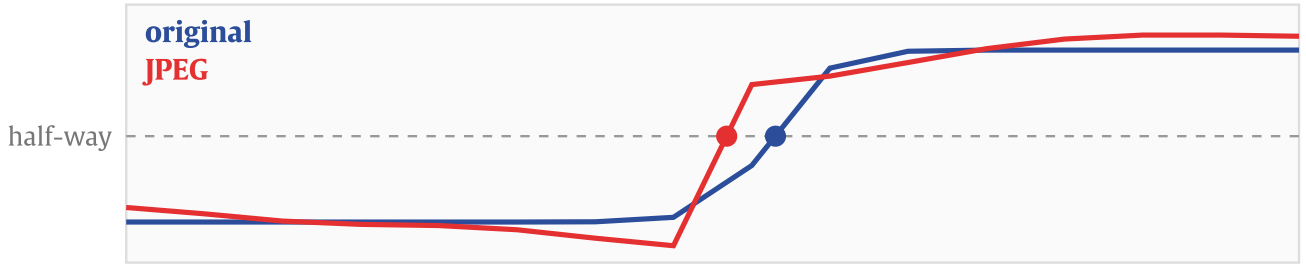
Lossless compression packs the data like a ZIP file: unpacked, every value is back. **Lossy** compression throws away what the eye notices least — which is not what a measurement needs least.

What lossy compression does to an edge

The same edge, saved without and with strong JPEG compression



original | after strong JPEG | where values changed (× 4)



grey levels along the green row: the edge (dot) moved by 0.62 pixel

- **How JPEG works:** it cuts the image into 8 × 8 blocks and stores each one as a sum of stripe patterns, keeping the coarse ones and rounding away the fine ones.
- **What you see:** the third image shows *where* the values changed (dark = large change, amplified four times). The flat areas are untouched; the error sits **along the edge** and at the block borders — exactly where we measure.
- **What it does to a measurement:** the edge position moves, by a different amount in every image — an uncontrolled error.

So

Measure on lossless files only (TIFF, PNG, raw). An 8-bit image of 5 megapixels is 5 MB: at 30 images a second, 150 MB/s — store the ones that matter, losslessly.

Images as quality records

An image of a rejected part is evidence — if it can be trusted

- **Save each image with its context:** date and time, camera settings, inspection program version, and the part's **serial number** — so the image can be traced to one part.
- **Keep:** every rejected part; a sample of accepted parts; the reference images used to check the system.
- **Store the original**, in a lossless file, and never edit it.
- **Keep it as long as** the quality system and the customer require — often years in automotive and medical supply.

> **ISO 9001, §7.5**
>
> Records that prove conformity must be kept and protected from changes. A stored inspection image is such a record.

ALCOA: five rules for records you can trust, from pharmaceutical good manufacturing practice (GMP) — good practice in any industry.

	Rule	For an inspection image
A	Attributable	who or which system made it is known
L	Legible	it can be opened and read for its whole life
C	Contemporaneous	recorded when the inspection happened
O	Original	the first file, not an edited copy
A	Accurate	it shows what was really there

Recap TAKEAWAYS

Digital images

1. **Why digital:** a digital image is data — computed on, calibrated, stored with its unit's serial number. It is what makes 100 % end-of-line inspection possible.
2. **Pixels:** each averages the light over its area. Two pixels per period, or detail aliases; 2–3 pixels across a defect.
3. **Grey levels:** 2^b levels, rounding error $\Delta/\sqrt{12}$. Saturation destroys information.
4. **Colour:** three grey images, because we have three cones. Change colour space to separate colour from light; measure colour with a calibrated set-up and CIELAB tolerances.
5. **Files:** lossless for anything measured or kept as a record; JPEG never. An image kept as evidence is an original, with its metadata.

Next
The sensor: how photons become these numbers, and where its noise comes from.

The hands-on lab
Your phone's photos as numbers: channels, grey levels, JPEG and noise.

Questions?

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github.com/LolloCappo/student-hub